YoungWoo Kim

+82-010-3312-7884 | aister9@naver.com | https://aister9.github.io/

aister9 | aister9

KOREATECH, Chungcheongnam-do, Republic of Korea

Education

- **KOREATECH(Korea University of Technology and Education)** *2013.03 - 2020.02*
  B.S. Computer engineering Chungcheongnam-do, Republic of Korea
  - GPA: 2.63/4.50
- **KOREATECH(Korea University of Technology and Education)** *2020.03 - 2022.02*
  M.S. Computer engineering Chungcheongnam-do, Republic of Korea
  - GPA: 4.25/4.50
  - Thesis: Real-Time Lens Distortion Algorithm on an Edge Device with GPU
- **KOREATECH(Korea University of Technology and Education)** *2022.02 - 2026.08*
  Ph.D. Computer engineering Chungcheongnam-do, Republic of Korea
  - GPA: 4.21/4.50
  - Dissertation (Expected): Hardware-Accelerated Ray Tracing-based acceleration for Hausdorff distance computation

Research Interest

- Spatial AI & Digital Twins
- High-Performance Computing (HPC)
- Physical Simulation and Proximity Queries
- Virtual and Augmented Reality (VR/AR)

Work and Teaching Experience

- **Rearch Assistant** *Chungcheongnam-do, Republic of Korea*
  SPIN(Spatial Intelligence) Lab. KOREATECH 2024 Spring - 2025 Fall
- **Part-time Lecturer** *Chungcheongnam-do, Republic of Korea*
  School of Computer Engineering, KOREATECH 2024 Spring - 2025 Fall
  - JAVA Programming (Fall 2025)
  - Multi-core programming (Spring 2024)
- **Teaching Assistant** *Chungcheongnam-do, Republic of Korea*
  School of Computer Engineering, KOREATECH 2020 Spring - 2023 Fall
  - Multi-core programming
  - Computer Graphics
  - Microprocessors
  - Algorithms
  - C Programming

PUBLICATIONS

GOOGLE SCHOLAR PROFILE

Journal Papers (SCI/SCIE)

[J.1] YoungWoo Kim, Jaehong Lee, Duksu Kim (2025). **RT-HDIST: Ray-Tracing Core-Based Hausdorff Distance Computation**. *Computer Graphics Forum*, Vol. 44, No. 7, e70229. (Pacific Graphics 2025 Journal Track).

[J.2] YoungWoo Kim, Sungmin Kwon, Duksu Kim (2025). **RTPD: Penetration Depth Calculation Using Hardware-Accelerated Ray-Tracing**. *The Visual Computer*, Vol. 41, No. 12, pp. 9885–9899.

[J.3] YoungWoo Kim, Hyeon-seok Yang, Duksu Kim (2022). **Real-Time Lens Distortion Algorithm on an Edge Device with GPU**. *IEEE Access*, Vol. 10, pp. 41748–41757.

[J.4] Jaehong Lee, You Chan No, YoungWoo Kim, Duksu Kim (2026). **A Large-Depth-Range Layer-Based Hologram Dataset for Machine Learning-Based 3D Computer-Generated Holography**. *Optics & Laser Technology*, Vol. 203, pp. 115636.

Conference Papers & Posters

[C.1] YoungWoo Kim, Duksu Kim (2026). **RT-VIS: Hardware Ray Tracing for Visibility-based Surface Reconstruction**. In *Proceedings of the Korea Computer Graphics Society Conference*, Accepted (To Appear).

[C.2] YoungWoo Kim, Duksu Kim (2023). **Penetration Depth Calculation Using Hardware Accelerated Ray Tracing Core**. In *Proceedings of the Korea Computer Graphics Society Conference*, pp. 77–78.

[P.1] Jaehong Lee, YoungWoo Kim, Duksu Kim (2025). **Color-Corrected Full Ray-Based Computer-Generated Holography**. In *Proceedings of SIGGRAPH Asia 2025 Posters*, pp. 1–2.

[P.2] YoungWoo Kim, Duksu Kim (2024). **Acceleration of 3D Surface Reconstruction Using Hardware-Accelerated Ray Tracing**. In *Proceedings of the Korea Computer Graphics Society Conference*, pp. 113–114.

[P.3] YoungWoo Kim, Duksu Kim (2022). **Real-Time Lens Distortion Algorithm on Embedded GPU Systems**. In *ACM SIGGRAPH 2022 Posters*, pp. 1–2.

[P.4] YoungWoo Kim, You Chan No, Sanghyun Lee, Duksu Kim (2022). **Simulation-Based Dataset Generation Method for Computer Vision Problems**. In *Proceedings of the Korea Computer Graphics Society Conference*, pp. 95–96.

RESEARCH GRANTS AND PROJECTS

- NRF, *Real-Time 3D Simulation via Ray-Tracing Core-Based Proximity Queries and 3D Reconstruction*, 2024–2026
- NRF, *High-Performance CGH Technology for Ultra-High-Resolution Hologram Generation*, 2021–2026
- IITP, *Holo-TV Core Technology Development for Hologram Video Services*, 2024
- IITP, *Development of Haptic Controller*, 2020–2022
- LINC+, *Development of an Autonomous Cart Platform with Scalability*, 2020–2022
- TIPA, *AI-Based Low-Cost 3D Robot Vision Platform Development* (with Hana Vision Tech Co.), 2020–2021
- NRF, *High-Performance Scientific Visualization Algorithms on Heterogeneous Parallel Computing Environments*, 2020–2021
- KOREATECH, *Visualization of 30 Years of KOREATECH Campus Evolution Using a Shape-Changing Display*, 2020–2021
- ETRI, *High-Performance Diffraction Computation for Accelerating Ultra-High-Resolution Hologram Generation*, 2021
- ETRI, *Large-Scale Matrix Processing Module for Accelerating Hologram Generation*, 2020
- Industry-funded projects with Samsung Electronics (2023) and MarkAny (2024) (confidential)

SKILLS

- **Programming:** C++, CUDA, Python, Java
- **Physical AI & Spatial Computing:** Digital Twins, 3D Reconstruction, Scene Understanding, Physical Simulation
- **GPU Computing:** CUDA, NVIDIA OptiX, RT Cores, Tensor Cores
- **Computer Graphics:** Point Clouds, Mesh Processing, Geometric Computing, Ray Tracing
- **Development Tools:** Git, Linux, Visual Studio, CMake

REFERENCES

1. **Prof. Duksu Kim**
   Department of Computer Science and Engineering
   KOREATECH (Korea University of Technology and Education)
   Relationship: Ph.D./M.S. Advisor